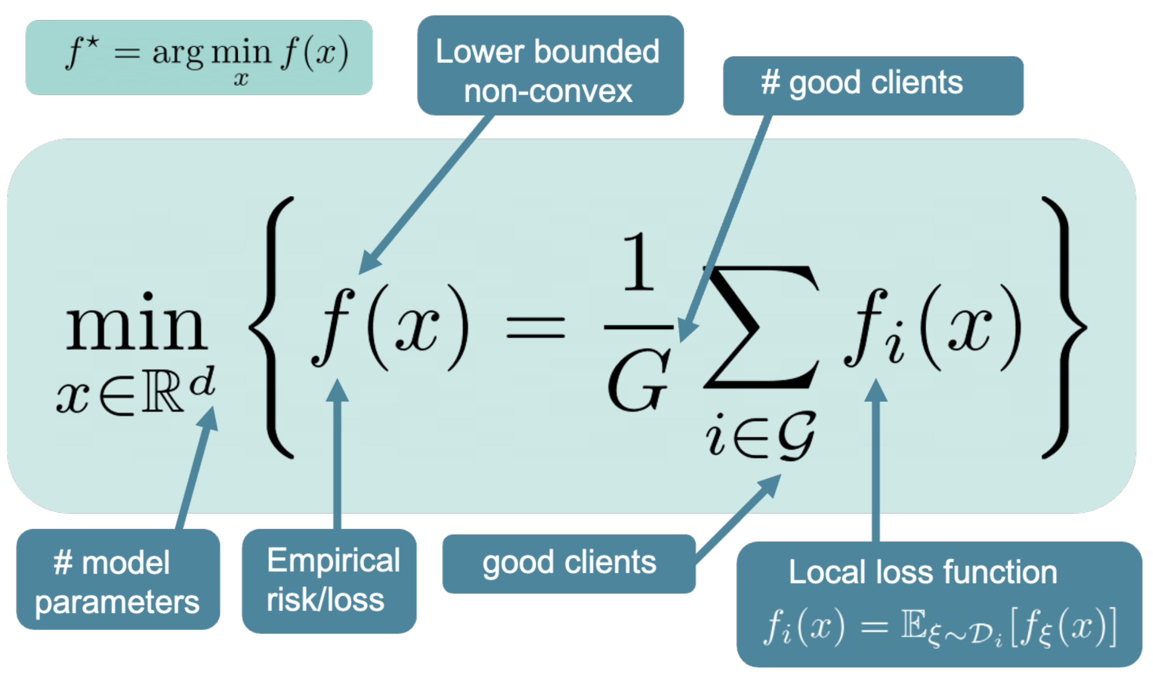
Byzantine-Robust and Differentially Private Federated Optimization under Weaker Assumptions

Rustem Islamov[1] Grigory Malinovsky[2] Alexander Gaponov[2] Aurelien Lucchi[1]
Peter Richtárik[2] Eduard Gorbunov[3]

[1]University of Basel [2]KAUST [3]MBZUAI

Problem Formulation



In our work, we use the following notation

- $\mathcal{G}$ is the set of regular clients with cardinality $G = |\mathcal{G}|$
- $\mathcal{B} = [n] \setminus \mathcal{G}$ is the set of Byzantine clients
- The fraction of Byzantine clients is bounded: $|\mathcal{B}|/n \le \delta_{\rm byz} < 1/2$

Our goal is to construct an algorithm that is both differentially private and Byzantine-robust *under standard assumptions.*

Differential Privacy and Robust Aggregation

Definition (Dwork et al., 2014). A randomized method $\mathcal{M}\colon \mathcal{D} \to \mathbb{R}$ satisfies (ε, δ)-Differential Privacy if for any two datasets $D, D' \in \mathcal{D}$ that differ in 1 sample and for any $S \subseteq \mathbb{R}$

$$\mathbb{P}(\mathcal{M}(D) \in S) \le e^{\varepsilon}\mathbb{P}(\mathcal{M}(D') \in \mathcal{S}) + \delta. \tag{1}$$

Definition (Allouah et al., 2023). The aggregation rule RAgg is said to be $(\delta_{\rm byz}, c)$-robust for some $c \ge 0$ if for any vectors $\{x_1, \dots, x_n\} \subseteq \mathbb{R}^d$ and any subset $S \subset [n]$ of size $n - |\mathcal{B}|$ the output $\hat{x} = \mathrm{RAgg}(x_1, \dots, x_n)$ satisfies

$$\|\hat{x} - \overline{x}\|^2 \le \frac{c\delta_{\rm byz}}{n - |\mathcal{B}|}\sum_{i\in S}\|x_i - \overline{x}\|^2 \tag{2}$$

where $\overline{x} = \frac{1}{n-|\mathcal{B}|}\sum_{i\in S} x_i$.

Proposed Algorithm

Algorithm 1: `Byz-Clip21-SGD2M`

Input: $x^0 \in \mathcal{X}$, $\gamma > 0$, $\beta, \hat{\beta} \in (0, 1]$, $v_i^0 = g_i^0 = m_i^0 \in \mathbb{R}^d$, $\tau > 0$, $\sigma_\omega^2 \ge 0$

For $t = 0, \dots, T-1$ **do**
- $x^{t+1} = x^t - \gamma g^t$
- **For** $i \in \mathcal{G}$ **do**
  - $v_i^{t+1} = (1-\beta)v_i^t + \beta\nabla f_i(x^{t+1}, \xi_i^{t+1})$
  - $\omega_i^{t+1} \sim \mathcal{N}(0, \sigma_\omega^2\mathbf{I})$
  - $c_i^{t+1} = \mathrm{clip}_\tau(v_i^{t+1} - g_i^t) + \omega_i^{t+1}$
  - $g_i^{t+1} = g_i^t + \hat{\beta}\mathrm{clip}_\tau(v_i^{t+1} - g_i^t)$
- **End For**
- **For** $i \in \mathcal{B}$ **do**
  - $c_i^{t+1} = (*)$ sends arbitrary vector
- **End For**
- $m_i^{t+1} = m_i^t + \hat{\beta}c_i^{t+1}$ $^{(*)}$ for $i \in [n]$
- $g^{t+1} = \mathrm{RAgg}(m_1^{t+1}, \dots, m_n^{t+1})$

End For

Table 1: Convergence guarantees of algorithms in the presence of Byzantine and/or DP adversaries. $\widetilde{\mathcal{O}}$ hides logarithmic and decaying-with-T terms. Notation: (s)CVX=(strongly) convex functions, nCVX=non-convex functions, DP=supports differential privacy, BR=supports Byzantine robustness, E=in-expectation analysis, P=high-probability analysis.

Method	Utility	DP	BR	Assumptions	Additional Comments	Type
`SoteriaFL` (Liu et al., 2022)	$\widetilde{\mathcal{O}}\left(\frac{\sqrt{d}}{\sqrt{G}\varepsilon}\right)$	✓	✗	M-bounded gradients	nCVX, local DP	E
α-`Norm` (Shulgin et al., 2025)	$\widetilde{\mathcal{O}}\left(\frac{\sqrt{d}}{\sqrt{G}\varepsilon} + R\right)^{(a)}$	✓	✗	L-smoothness Full-batch gradients	nCVX, local DP	E
`Clip21-SGD2M` (Islamov et al., 2025)	$\widetilde{\mathcal{O}}\left(\frac{\sqrt{d}}{\sqrt{G}\varepsilon}\right)^{(b)}$	✓	✗	L-smoothness σ-sub-Gaussian noise	nCVX, local DP	P
`Byz-SGDM` (Karimireddy et al., 2020)	$\widetilde{\mathcal{O}}\left(\delta_{\rm byz}\zeta^2\right)^{(c)}$	✗	✓	L-smoothness Bounded variance	nCVX	E
`Robust D-SHB` (Allouah et al., 2023)	$\widetilde{\mathcal{O}}\left(\delta_{\rm byz}M^2\right)$	✗	✓	L-smoothness Full-batch gradients	nCVX CVX	E
`Byz-VR-MARINA` (Gorbunov et al., 2022)	$\widetilde{\mathcal{O}}\left(\delta_{\rm byz}\zeta^2\right)$	✗	✓	L-smoothness Local functions are finite-sums	nCVX	E
`Safe-DSHB` (Allouah et al., 2023)	$\widetilde{\mathcal{O}}\left(\frac{d}{G\varepsilon^2} + \frac{\delta_{\rm byz}}{\varepsilon^2} + \delta_{\rm byz}M^2\right)$	✓	✓	M-bounded gradients	sCVX, local DP	E
`DP-BREM` (Gu et al., 2025)	$\widetilde{\mathcal{O}}\left(\delta_{\rm byz}\zeta^2 + \delta_{\rm byz}\frac{\sqrt{d}}{\sqrt{G}\varepsilon}\right)^{(d)}$	✓	✓	L-smoothness Momentum buffers are i.i.d	nCVX, central DP	E
`P&S Learning` (Xiang et al., 2023)	N/A$^{(e)}$	✓	✓	L-smoothness $\nabla f_i(x) - \nabla f_i(x^\star)$ are sub-Gaussian Access to external non-private dataset	sCVX, local DP	P
`Byz-Clip21-SGD2M` **(Ours)**	$\widetilde{\mathcal{O}}\left(R + R^{2/3}\right)^{(b,f)}$ $R = \frac{\sqrt{d}}{\sqrt{G}\varepsilon} + \frac{\sqrt{\delta_{\rm byz}d}}{\varepsilon} + \sqrt{\delta_{\rm byz}\mathcal{N}}$	✓	✓	L-smoothness σ-sub-Gaussian noise	nCVX, local DP	P

(a) Here $R := \max_{i\in\mathcal{G}}\|\nabla f_i(x^0) - g_i^0\|$. (b) Derived under the assumption $\sqrt{d}/\sqrt{G}\varepsilon \ge 1$.
(c) ζ^2 denotes the constant in the gradient dissimilarity assumption. (d) The provided rate contradicts the lower bound due to the unrealistic assumption that momentum buffers are i.i.d. distributed.
(e) The authors do not provide explicit bounds. (f) In the general case, $\mathcal{N} = LF^0 + \zeta^2 + \sigma(\sqrt{LF^0} + \zeta)$, where $F^0 = f(x^0) - f^\star$. In the presence of Byzantine adversaries only, $\mathcal{N} = \sqrt{\delta_{\rm byz}}\zeta^2$.

Experiments

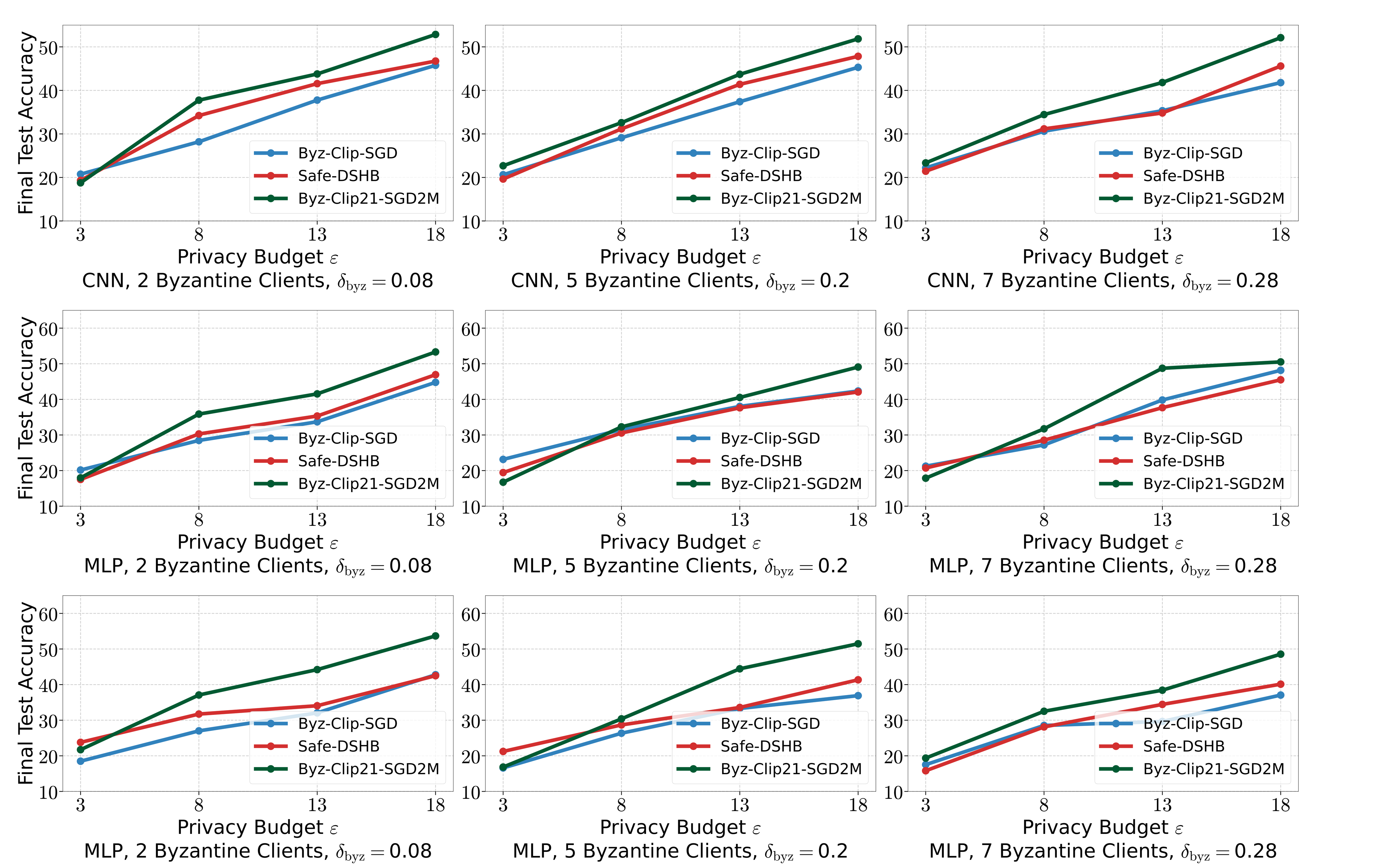


Figure 1: Performance of `Byz-Clip21-SGD2M`, `Byz-Clip-SGD`, and `Safe-DSHB` when training CNN and MLP models on the MNIST dataset for different numbers of Byzantine clients and privacy budgets, when Byzantine clients use `IPM` attack (two top lines) and `Label Flip` (bottom line).

Theoretical Analysis

Assumptions.

(A1) f_i is L-smooth for all $i \in [n]$;
(A2) $\nabla f_i(x, \xi)$ is unbiased estimator of $\nabla f_i(x)$ and σ-sub-Gaussian for all $i \in [n]$;
(A3) $f^\star = \inf_{x\in\mathbb{R}^d} f(x) > -\infty$;
(A4) f_i's satisfy $\frac{1}{G}\sum_{i\in\mathcal{G}}\|\nabla f_i(x) - \nabla f(x)\|^2 \le \zeta^2$;

The analysis uses the Lyapunov function

$$\Phi^t := f(x^t) - f^\star + \frac{8\gamma\beta}{\hat{\beta}^2\eta^2}\frac{1}{G}\sum_{i\in\mathcal{G}}\|v_i^t - \nabla f_i(x^t)\|^2 + \frac{2\gamma}{\hat{\beta}\eta}\frac{1}{G}\sum_{i\in\mathcal{G}}\|g_i^t - v_i^t\|^2 + \frac{2\gamma}{\beta}\|\overline{v}^t - \nabla f(x^t)\|^2.$$

Theorem (Simplified)

The iterates of `Byz-Clip21-SGD2M` run with DP noise variance $\sigma_\omega^2 > 0$ and $(c, \delta_{\rm byz})$-robust aggregator satisfy with probability at least $1 - \alpha$ that

$$\frac{1}{T}\sum_{t=0}^{T-1}\|\nabla f(x^t)\|^2 \le \widetilde{\mathcal{O}}\left(\Delta_0(R + R^{2/3})\right),$$

where

$$R = \frac{\sqrt{d}\sigma_\omega}{\sqrt{GT}\tau} + \frac{\sqrt{c\delta_{\rm byz}d}\sigma_\omega}{\sqrt{T}\tau} + \sqrt{c\delta_{\rm byz}}, \quad F^0 := f(x^0) - f^\star,$$

$$\Delta_0 := \sqrt{L\Delta}(\sqrt{L\Delta} + \widetilde{B}_{\rm init} + \sigma) = \widetilde{\mathcal{O}}(LF^0 + \zeta^2 + \sigma(\sqrt{LF^0} + \zeta)).$$

Corollary (Simplified)

With $\sigma_\omega = \Theta\left(\frac{\tau}{\varepsilon}\sqrt{T\log(\frac{T}{\delta})\log(\frac{1}{\delta})}\right)$, all iterations of `Byz-Clip21-SGD2M` satisfy local (ε, δ)-DP with robustness-privacy-utility trade-off bounded with probability at least $1-\alpha$ as in the theorem above with

$$R = \frac{\sqrt{d}}{\sqrt{G}\varepsilon} + \sqrt{c\delta_{\rm byz}}\frac{\sqrt{d}}{\varepsilon} + \sqrt{c\delta_{\rm byz}}.$$

Special Cases.

- if $\varepsilon = \delta_{\rm byz} = 0$, the rate is $\widetilde{\mathcal{O}}(1/T + \sigma/\sqrt{T})$
- if $\delta_{\rm byz} > 0$, the utility $\Delta_0 \cdot \widetilde{\mathcal{O}}\left(\sqrt{d}/\sqrt{G}\varepsilon + \left(\sqrt{d}/\sqrt{G}\varepsilon\right)^{2/3}\right)$, which recovers the results of (Islamov et al., 2025);
- if $\varepsilon = 0$, the utility is $\widetilde{\mathcal{O}}(1/T + \sigma/\sqrt{T} + c\delta_{\rm byz}\zeta^2)$, which recovers the results of (Karimireddy et al., 2020).

Main Challenges in the Analysis.

- RAgg induces a bias between g^t and $\frac{1}{G}\sum_{i\in\mathcal{G}} m_i^t$
- This bias scales multiplicatively with σ_ω and $c\delta_{\rm byz}$
- No bounded gradients assumption $\Longrightarrow$ careful analysis is needed
- To address this, we treat clipping as a contractive compressor with *input-dependent contraction factor* $\Longrightarrow$ high-probability analysis
- We prove by induction that the inputs to the clipping operator remain bounded with high probability
- Obtaining polylog-dependence on the failure probability α hinges on a careful decomposition of the noise terms, each controlled via concentration inequalities